Calculator Model / No.

Class	Reg Number

Candidate's Name: ______________________________

MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

CHEMISTRY 9647/01

Paper 1 Multiple-Choice Questions

23 September 2015

1 hour

Additional Materials: Data Booklet and OMR answer sheet

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

There are **forty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

<u>Use of OMR Answer Sheet</u>

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the ***OMR sheet***, please follow the given examples:

If your register number is **1**, then shade **<u>01</u>** in the index number column.

If your register number is **21**, then shade **<u>21</u>** in the index number column.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 A sample of 10 cm^3 of methanol was burnt in 50 cm^3 of oxygen. After measuring the volume of gas remaining, the product was treated with an excess of aqueous sodium hydroxide and the volume of gas measured again. All measurements were made at temperature and pressure in which methanol was gaseous and water was liquid.

What were the measured volumes?

	Volume of O_2 unreacted / cm^3	Volume of residual gas after burning / cm^3	Volume of residual gas after treated with NaOH / cm^3
A	15	45	35
B	20	40	30
C	30	40	30
D	35	45	35

2 In an experiment, H_2S was reacted with 24.00 cm^3 of 0.15 mol dm^{-3} of an unknown bromate ion, BrO_x^- in acidic medium to form sulfur of mass 0.289 g as well as bromine solution, Br_2.

What is the value of ***x*** in BrO_x^-?

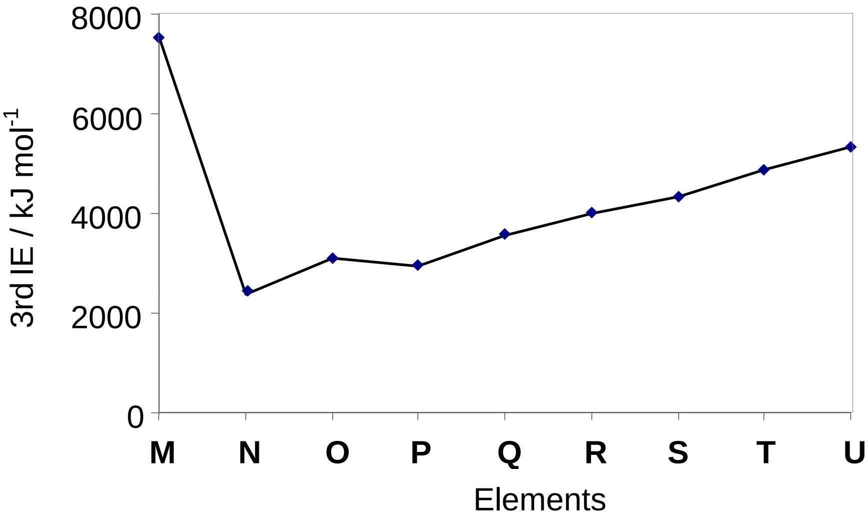
A 2

B 3

C 4

D 5

3 The graph below shows the variation in the **third** ionisation energies for the consecutive elements **M** to **U** in the Periodic Table, all with proton number below 20. The symbols **M** to **U** do not represent the notation of the actual elements.



What can be deduced from the above graph?

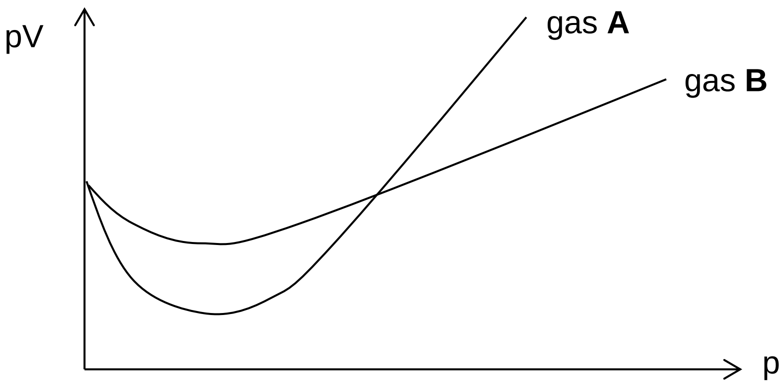
A **M** has a noble gas configuration.

B **M** reacts with **R** to form the compound **M**$\mathbf{R_2}$.

C The atomic radius of **T** is smaller than that of **S**.

D The decrease in 3rd I.E from **O** to **P** is due to inter-electron repulsion.

4 The value of pV is plotted against p for two gases, **A** and **B**, where p is the pressure and V is the volume of the gas.



Which of the following could **not** be the identities of the gases?

	Gas **A**	Gas **B**
A	CO_2 at 25 ^oC	SO_2 at 25 ^oC
B	N_2 at 25 ^oC	N_2 at 100 ^oC
C	NH_3 at 25 ^oC	H_2 at 25 ^oC
D	HF at 25 ^oC	HC*l* at 25 ^oC

5 Which combination of acid and base would result in the highest rise in temperature?

A 25 cm^3 of 1 mol dm^{-3} of $Ba(OH)_2$ and 25 cm^3 of 1 mol dm^{-3} of H_2SO_4

B 50 cm^3 of 1 mol dm^{-3} of NaOH and 25 cm^3 of 1 mol dm^{-3} of H_2SO_4

C 50 cm^3 of 1 mol dm^{-3} of NaOH and 50 cm^3 of 1 mol dm^{-3} of H_3PO_4

D 75 cm^3 of 1 mol dm^{-3} of $Ba(OH)_2$ and 75 cm^3 of 1 mol dm^{-3} of HCl

6 Which of the following has a positive enthalpy change of reaction?

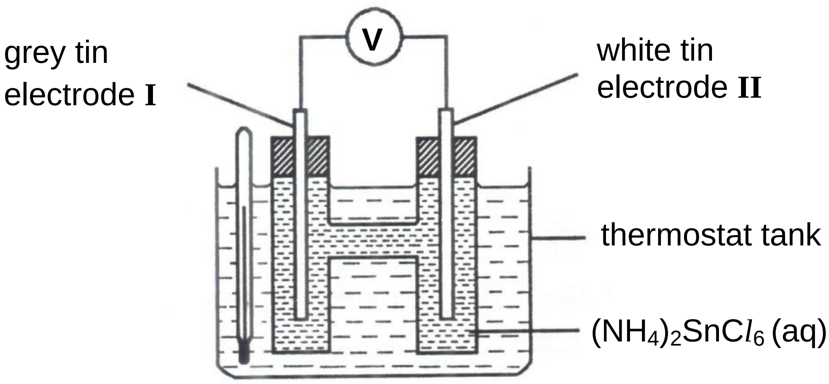
A dimerisation of NO_2 radical

B dissolving ammonium chloride in water

C formation of carbon dioxide gas

D formation of Al_2O_3 solid from gaseous Al^{3+} and O^{2-}

7 The diagram shows an apparatus used to determine the temperature at which white tin and grey tin are in equilibrium. This is found to be 15 ºC. Above 15 ºC, grey tin from electrode **I** dissolves and is deposited on electrode **II** as white tin.



Which of the following are the features of the cell obtained?

A At 15 ^oC, there is current flow.

B Above 15 ^oC, electrode **II** acts as anode.

C Increasing the size of the grey tin electrode will not change the cell potential.

D Porous grey and white tin electrodes can be used to increase the cell potential.

8 A molten chloride of a Group I metal, **M**C*l*, and a molten chloride of a Group II metal, **N**Cl_2, is separately electrolysed using the same current for the same period of time.

Which of the following statements about the experiment is correct?

A Equal number of moles of **M** and **N** are deposited at cathode.

B The mass of **N** deposited is twice that of the mass of **M** deposited.

C The mass of elements **M** and **N** deposited is independent of the duration of the electrolysis.

D Equal volume of Cl_2 gas is evolved by **M**C*l* and **N**Cl_2 within a given period of time at the same temperature.

9 One of the key production stages in the Contact Process is the production of sulfur trioxide.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

The rate constants of the forward and backward reactions are given as k_1 and k_{-1} respectively.

What happens to k_1, k_{-1}, K_c and the equilibrium position if an inert gas is introduced into the reaction vessel at constant volume?

	k_1	k_{-1}	K_c	*equilibirum position*
A	unchanged	unchanged	unchanged	unchanged
B	unchanged	unchanged	unchanged	shifts to right
C	unchanged	unchanged	unchanged	shifts to left
D	decreases	increases	decreases	shifts to left

10 In an experiment, 1.00 x 10^{-2} mol of silver nitrate, $AgNO_3$, and 9.00 x 10^{-3} mol of calcium nitrate, $Ca(NO_3)_2$, are dissolved in a solution containing 0.100 mol dm^{-3} of sulfuric acid, H_2SO_4.

The solubility product of Ag_2SO_4 is 1.20 x 10^{-5} mol^3 dm^{-9} and that of $CaSO_4$, is 7.10 x 10^{-5} mol^2 dm^{-6}.

Which statement is correct?

A No precipitate is formed.

B Only $CaSO_4$ is precipitated.

C $CaSO_4$ is precipitated followed by Ag_2SO_4.

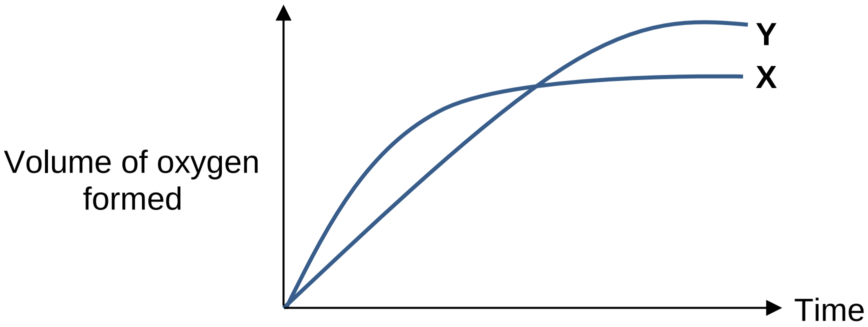
D Ag_2SO_4 is precipitated followed by $CaSO_4$.

11 To separate same volumes of 0.0100 mol dm^{-3} of a strong monobasic acid **A** and 0.100 mol dm^{-3} of a weak monobasic acid **B**, an equal volume of distilled water was added. The K_a of acid **B** is 1.75 x 10^{-5} mol dm^{-3}.

What is the pH of the resulting solutions of acid **A** and acid **B**?

	acid **A**	acid **B**
A	2.00	2.88
B	2.00	3.03
C	2.30	2.88
D	2.30	3.03

12 In the diagram, curve **X** was obtained by observing the decomposition of 100 cm^3 of 1 mol dm^{-3} hydrogen peroxide with manganese(IV) oxide as catalyst.



Which alteration to the original experiment conditions would produce curve **Y**?

A adding 10 cm^3 of 0.1 mol dm^{-3} of hydrogen peroxide

B using more manganese(IV) oxide

C lowering the temperature

D adding water

13 **W** is a Group II element that is diagonal to aluminium in the Periodic Table.

Which of the following would **not** be a correct prediction of **W** or its compounds?

A The chloride of **W** exists as simple molecules in gaseous state.

B The oxide of **W** forms a colourless solution with aqueous sodium hydroxide.

C The oxide of **W** is insoluble in water.

D The oxide of **W** conducts electricity.

14 What is the flame colour when barium is burnt in air?

A apple green

B brick red

C bright white

D crimson

15 Due to its radioactive nature, the properties of astatine, At, have to be estimated based on its position in the Periodic Table.

Which of the predictions concerning At or its compounds is correct?

A Hydrogen astatide has a higher decomposition temperature than hydrogen bromide.

B Astatine is a light coloured solid at room temperature.

C Silver astatine is soluble in concentrated ammonia.

D Iodine is a stronger oxidising agent than astatine.

16 The electronic configuration of a transition element is shown below.

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$$

What oxidation state for this element does **not** occur in its compounds?

A 0

B +2

C +3

D +7

17 *Use of the Data Booklet is relevant to this question.*

Which of the following statement about iron complexes is **not** correct?

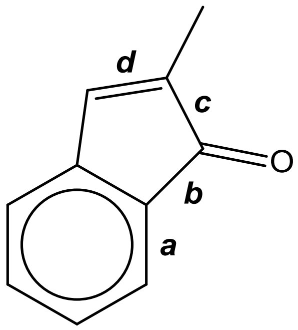
A $[Fe(CN)_6]^{3-}$ is less stable than $[Fe(CN)_6]^{4-}$.

B A precipitate and a gas is formed when Na_2CO_3 is added to a solution containing Fe^{3+} ions.

C The addition of aqueous H_2O_2 to $Fe(OH)_2$ results in the formation of $Fe(OH)_3$.

D CN^- ligands in yellow $[Fe(CN)_6]^{4-}$ complex causes a smaller d–orbital splitting compared to H_2O ligands in pale green $[Fe(H_2O)_6]^{2+}$ complex.

18 Four carbon-carbon bonds are labelled in the diagram below.



Which bond formation involves sp^2–sp^2 overlap?

A **a** and **d** only

B **a** and **b** only

C **a**, **b** and **d** only

D **a**, **b**, **c** and **d**

19 Ethane was mixed with bromine gas in a closed vessel. When exposed to ultraviolet light, a reaction occurred and a mixture of products was formed.

Which of the following statements shows that a free radical reaction has occurred?

A A trace amount of butane was found in the mixture.

B Bromoethane is a major product of the reaction.

C The reddish-brown bromine gas was decolourised.

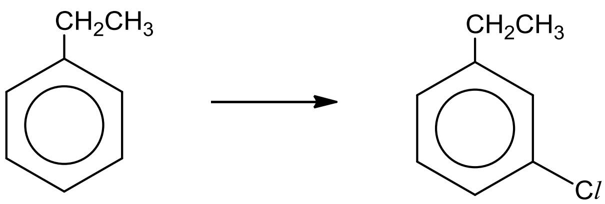
D Ultraviolet light is required for the reaction.

20 Which transformation can be readily achieved by only one substitution reaction?

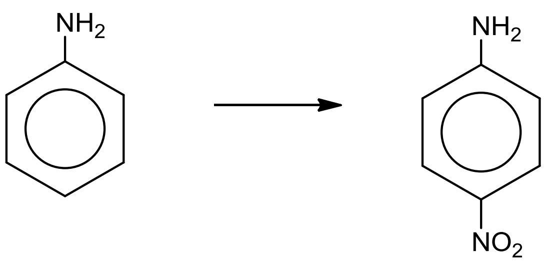
A $CH_3CH_2CH_2CH_2F \longrightarrow CH_3CH_2CH_2CH_2NH_2$

B $CH_3CH_2CH{=}CH_2 \longrightarrow CH_3CH_2CH{=}CHBr$

C



D



[Turn over

21 The use of chlorofluoroalkanes, CFCs, has been heavily regulated because of their destructive effects on the ozone layer. High-energy radiation in the stratosphere produces free-radicals from CFCs.

Given that the attachment of alkyl groups stabilise the carbon radical, which free-radical is most likely to result from the irradiation of $CF_3CFClCF_2Cl$?

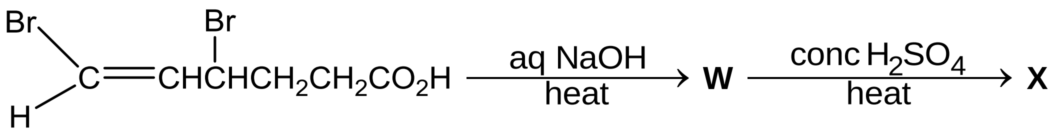
A $CF_3\dot{C}ClCF_2Cl$

B $CF_3\dot{C}FCF_2Cl$

C $CF_3CFCl\dot{C}FCl$

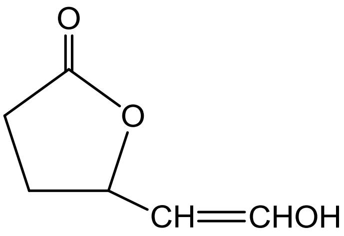
D $CF_3CFCl\dot{C}F_2$

22 The reaction scheme outlines a two−step synthesis to form **X**.

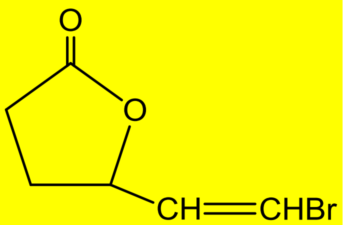


Which compound is **X**?

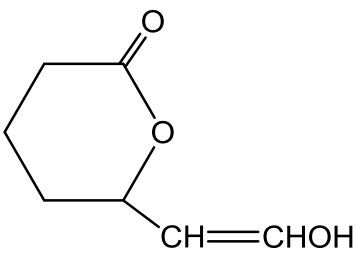
A



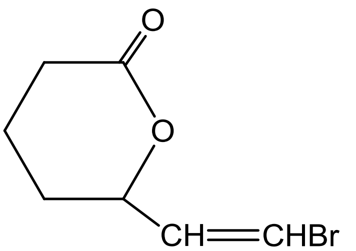
B



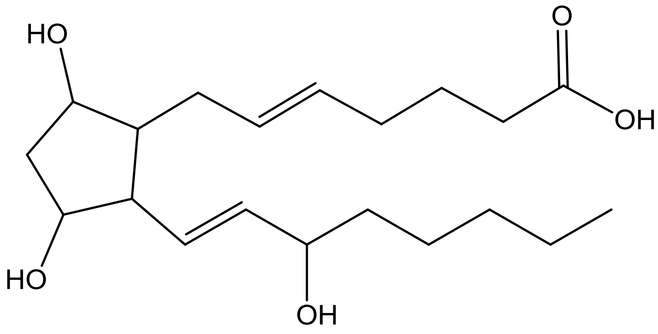
C



D



23 *Prostaglandin* is a naturally occurring prostaglandin used in medicine to induce labour.



prostaglandin

What reacts exactly with one mol of *prostaglandin* under suitable conditions?

A 3 moles of NaOH(aq)

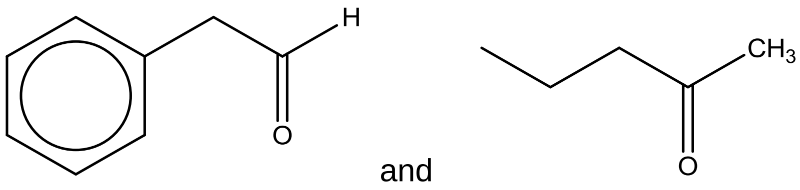
B 3 moles of Br_2(aq)

C 4 moles of PCl_5(s)

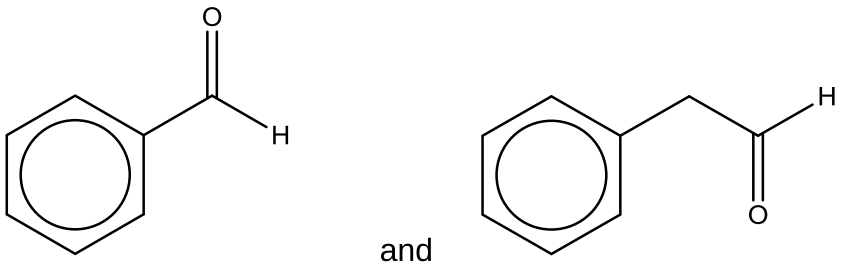
D 4 moles of C_6H_5COCl(s)

24 Which of the following pairs of compounds **cannot** be differentiated by heating with an alkaline solution of Cu^{2+} ions?

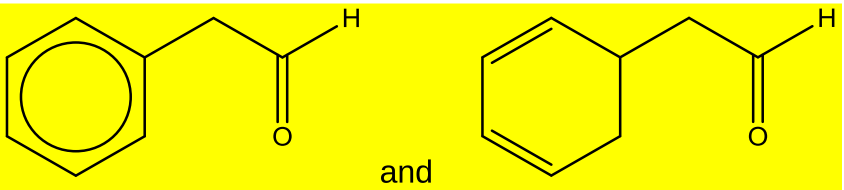
A



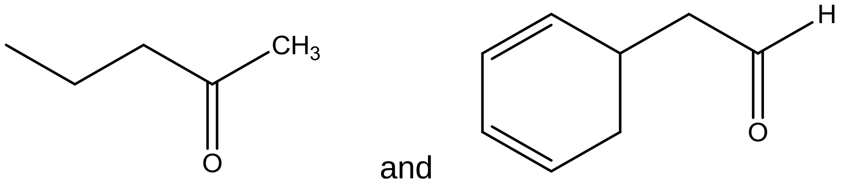
B



C

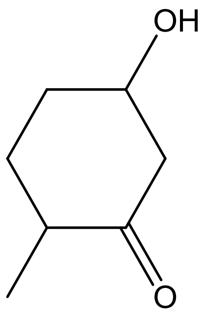


D

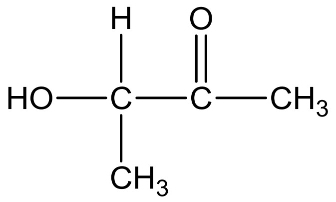


25 Which of the following compounds, upon reaction with hot acidified $KMnO_4$ followed by H_2 with Pt catalyst, can form a product that has four optical isomers?

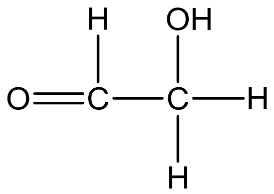
A



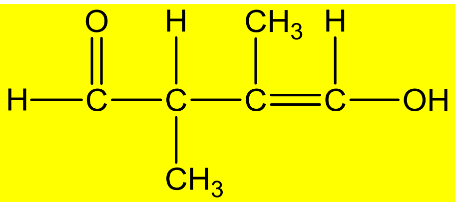
B



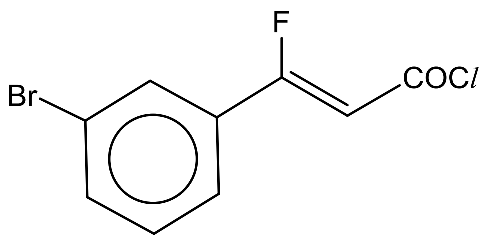
C



D



26 How many moles of silver halide would be produced when one mole of the following compound is reacted with aqueous silver nitrate at room temperature?



A 0 **B** 1 **C** 2 **D** 3

27 In which sequence is it correctly stated that the value of pK_a decreases continuously?

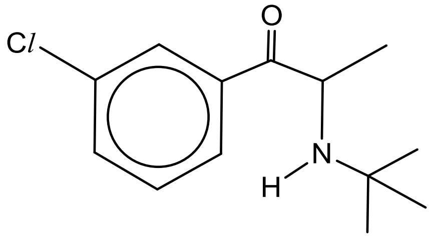
A $CH_3CO_2H > CCl_3CO_2H > C_2H_5OH > C_6H_5OH$

B $CCl_3CO_2H > CH_3CO_2H > C_2H_5OH > C_6H_5OH$

C $C_2H_5OH > C_6H_5OH > CH_3CO_2H > CCl_3CO_2H$

D $C_6H_5OH > C_2H_5OH > CH_3CO_2H > CCl_3CO_2H$

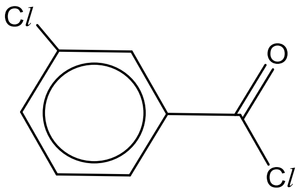
28 *Bupropion* is an anti-depressant used in the treatment of hyperactive children and adults.

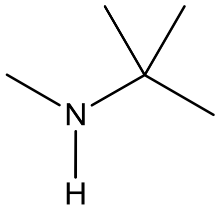


Bupropion

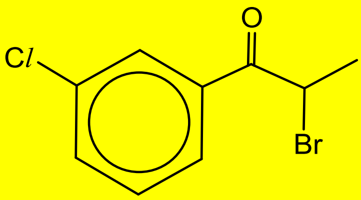
Which pair of compounds would produce *Bupropion* in a one-step reaction when reacted together under suitable conditions?

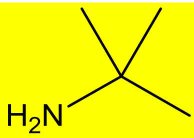
A



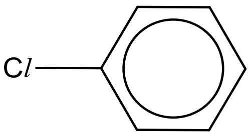


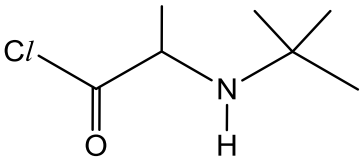
B



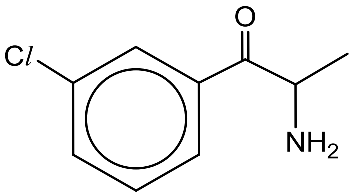


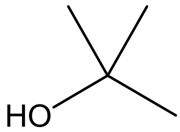
C





D





29 Phenylamine can be made from nitrobenzene in the laboratory using the method given below.

4.84 g of nitrobenzene and 10 g of tin are placed into a round bottom flask which is heated under reflux for 30 minutes. 20 cm^3 of concentrated hydrochloric acid is added dropwise to the flask. Once the reaction mixture is cooled, excess concentrated sodium hydroxide solution is added.

Which of the following statement does **not** explain why excess concentrated sodium hydroxide must be added to the reaction mixture?

A To generate the product phenylamine from the salt formed.

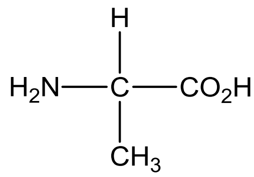
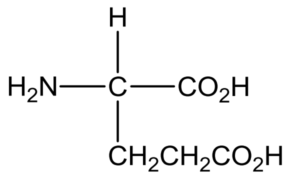
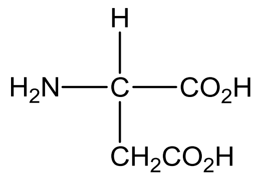
B To remove the tin(II) ions formed from the reduction of nitrobenzene.

C To separate the phenylamine product by forming two immiscible layers.

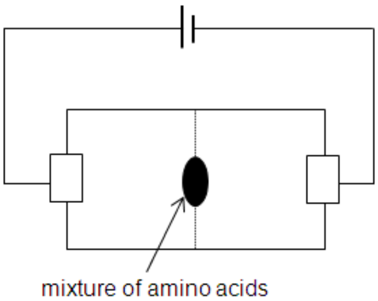
D To react with the side products formed from the reduction of nitrobenzene.

[Turn over

30 The table below gives some information pertaining to three amino acids.

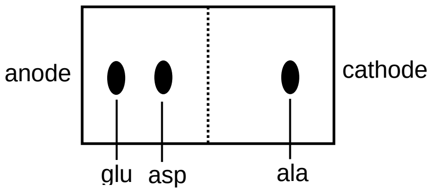
Amino acid	**Structure**	**Isoelectric point**
alanine (ala)		6.0
glutamic acid (glu)		3.1
aspartic acid (asp)		2.9

A mixture of these three amino acids can be separated by electrophoresis.

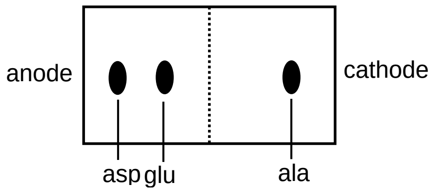


Which of the following diagram shows the result of the separation of the amino acid mixture at pH 4.5?

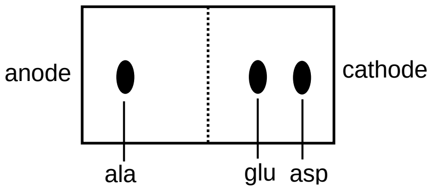
A



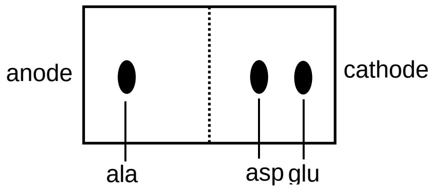
B



C



D



Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 Which of the following pairs have the same bond angle?

1 NO_3^-, CO_3^{2-}

2 NO_2, SO_2

3 XeF_4, CF_4

32 Which of the following can be explained by hydrogen bonding?

1 the existence of HF_2^-

2 the lower density of ice as compared to water

3 the lower acidity of aqueous HF in water as compared to aqueous HBr

33 Which of the following equation correctly defines the enthalpy change of formation of a compound?

1 $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$

2 $Na(g) + Cl(g) \longrightarrow NaCl(s)$

3 $2Cr(s) + \frac{3}{2}O_2(l) \longrightarrow Cr_2O_3(s)$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

34 The kinetics of the reaction below was studied using the method of initial rates.

$$I^-(aq) + ClO^-(aq) \xrightarrow{OH^-} IO^-(aq) + Cl^-(aq)$$

Experiment	Initial $[I^-]$ / mol dm^{-3}	Initial $[ClO^-]$ / mol dm^{-3}	Initial $[OH^-]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.0013	0.012	0.100	9.4×10^{-3}
2	0.0026	0.012	0.100	18.7×10^{-3}
3	0.0013	0.018	0.100	14.0×10^{-3}
4	0.0026	0.018	0.050	56.1×10^{-3}

Which conclusions can be drawn about the reaction?

1 OH^- is acting as a catalyst.

2 The overall order of reaction is 1.

3 The rate constant is 59.9 s^{-1}.

35 Magnesium is a metal and chlorine is a non-metal. Which of the following conclusions can be drawn from this information?

1 Magnesium and chlorine has the same number of quantum shells of electrons.

2 Magnesium is a stronger reducing agent than chlorine.

3 Magnesium is more electropositive than chlorine.

36 Which of the following reactions will give bromine as the product?

1 Sodium bromide is shaken with aqueous chlorine.

2 Hydrogen bromide is bubbled through concentrated sulfuric acid.

3 Sodium bromide is reacted with potassium thiosulfate.

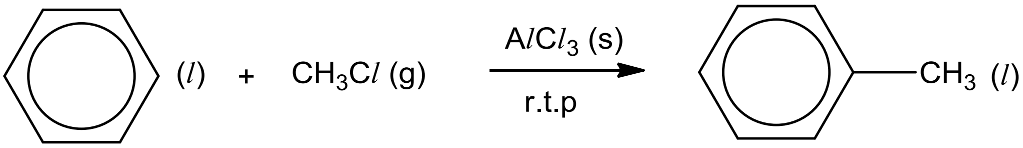
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

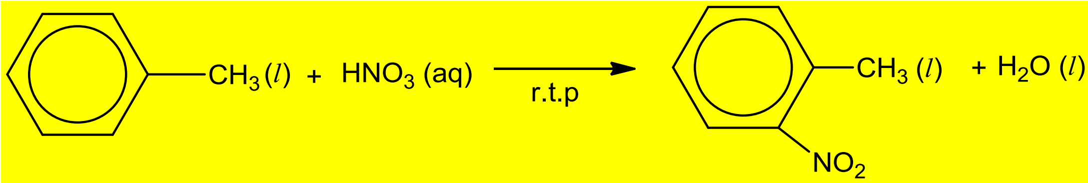
No other combination of statements is used as a correct response.

37 Which of the following transformation has a set of conditions that is **not** correct?

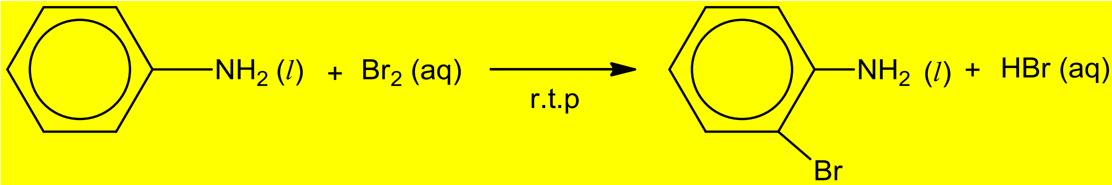
1



2

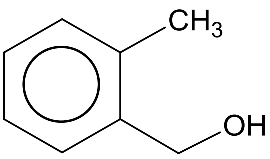


3



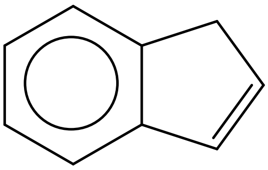
38 Which of the following organic compounds will produce benzene–1,2–dicarboxylic acid when subjected to the reagents and conditions as specified?

1



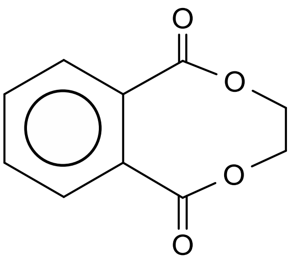
, $K_2Cr_2O_7$ /H^+, heat

2



, $KMnO_4$/H^+, heat

3



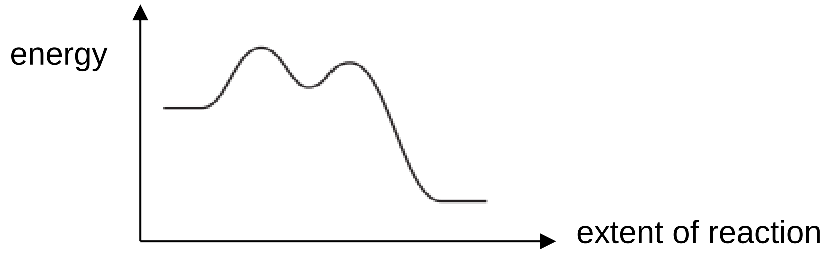
, H^+(aq), heat

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

39 The following sketch depicts a reaction pathway.



Which reactions would have this profile?

1 $C(CH_3)_3CH_2Br + NaOH \longrightarrow C(CH_3)_3CH_2OH + NaBr$

2 $C(CH_3)_2BrCH_3 + KCN \longrightarrow C(CH_3)_2(CN)CH_3 + KBr$

3 $C(CH_3)_3C(CH_3)_2Cl + 2NH_3 \longrightarrow C(CH_3)_3C(CH_3)_2NH_2 + NH_4Cl$

40 When one mole of compound **P** is reacted with excess sodium metal, exactly one mole of hydrogen gas is evolved. When one mole of compound **P** is reacted with excess sodium carbonate, exactly one mole of carbon dioxide gas is evolved.

Which of the following could be **P**?

1 $HO_2CCH_2CO_2H$

2 $HOCH_2CH_2CO_2H$

3 $HOC_6H_4CO_2H$

END OF PAPER